

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

Practical – IV

Roll No.: T040	Name: Omkar Prakash Sawant
Class: TY.IT	Batch: 1
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Bayes' Rule Application

1. In a college:

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.

Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

Code:

```
P_Python = 0.20
P_Java = 0.80
P_Placed_Python = 0.90
P_Placed_Java = 0.40
P_Placed = (P_Placed_Python * P_Python) + (P_Placed_Java * P_Java)
P_Python_Placed = (P_Placed_Python * P_Python) / P_Placed
print("Probability that the student knows Python:", P_Python)
print("Probability that the student knows Java:", P_Java)
print("Probability that a placed student knows Python:")
print(round(P_Python_Placed * 100, 2), "%")
```

Output:

```
Probability that the student knows Python: 0.2
Probability that the student knows Java: 0.8
Probability that a placed student knows Python:
36.0 %
```

2. In a college:

- 20% of students know Python.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

A randomly selected student is placed.

Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

Code:

```
P_Python = 0.20
P_Java = 0.80
P_Placed_Python = 0.90
P_Placed_Java = 0.40
P_Placed = (P_Placed_Python * P_Python) + (P_Placed_Java * P_Java)
P_Python_Placed = (P_Placed_Python * P_Python) / P_Placed
print("Probability that the student knows Python:", P_Python)
print("Probability that the student knows Java:", P_Java)
print("Probability that a placed student knows Python:")
print(round(P_Python_Placed * 100, 2), "%")
```

Output:

```
Probability that the student knows Python: 0.2
Probability that the student knows Java: 0.8
Probability that a placed student knows Python:
36.0 %
```

1. Probability that the student knows Python

Given in the question:

$P(\text{Python}) = 20\% = 0.20$

Answer: 0.20 (20%)

2. Probability that the student knows Java

Given in the question:

$P(\text{Java}) = 80\% = 0.80$

Answer: 0.80 (80%)

3. Probability that a placed student knows Python

Using Bayes' Theorem:

$$P(\text{Python} | \text{Placed}) = \frac{P(\text{Placed} | \text{Python}) \times P(\text{Python})}{P(\text{Placed})}$$
$$P(\text{Placed}) = (0.90 \times 0.20) + (0.40 \times 0.80) = 0.18 + 0.32 = 0.50$$

First, calculate the total probability of getting placed:

$$P(\text{Placed}) = (0.90 \times 0.20) + (0.40 \times 0.80) = 0.18 + 0.32 = 0.50$$

Now apply Bayes' Theorem:

$$P(\text{Python} | \text{Placed}) = \frac{0.90 \times 0.20}{0.50} = \frac{0.18}{0.50} = 0.36$$

Answer: 0.36 (36%)

Q.2 Taxi Colour Problem (Human Answer)

Given:

- Green taxis = **85%**
- Blue taxis = **15%**
- Witness identifies correctly = **80%**
- Witness identifies incorrectly = **20%**

Find the probability that the taxi is actually Blue if the witness says it is Blue.

Using Bayes' Theorem:

$$P(\text{Blue} | \text{Says Blue}) = \frac{P(\text{Says Blue} | \text{Blue}) \times P(\text{Blue})}{P(\text{Says Blue})}$$
$$P(\text{Says Blue}) = (0.80 \times 0.15) + (0.20 \times 0.85) = 0.12 + 0.17 = 0.29$$

First, calculate the probability that the witness says "Blue":

$$P(\text{Says Blue}) = (0.80 \times 0.15) + (0.20 \times 0.85) = 0.12 + 0.17 = 0.29$$

Now apply Bayes' Theorem:

$$P(\text{Blue} | \text{Says Blue}) = \frac{0.80 \times 0.15}{0.29} = \frac{0.12}{0.29} \approx 0.4138$$

Answer: 0.4138 (41.38%)

Final Answers

Q.1

- Probability that the student knows Python = **20%**
- Probability that the student knows Java = **80%**
- Probability that a placed student knows Python = **36%**

Q.2

- Probability that the taxi is actually Blue when the witness says Blue = **41.38%**